

In the past, when companies needed servers, they would order physical servers based on their specific resource requirements like CPU and RAM. These servers would be delivered physically, and then the company would implement their applications on them to serve public websites. However, in this setup, resource utilization could be inefficient—sometimes traffic would be low, leading to wasted resources. This inefficiency led to the development of a new technology: virtualization.

Virtualization still involves physical servers, but it introduces a layer where an operating system is installed, followed by a hypervisor—a software that allows the division of resources into multiple virtual machines (VMs). These VMs act like individual machines but are actually segregated portions of the physical server, allowing for more efficient use of resources.

There are many virtualization software options available, like VMware Workstation, which allows you to logically divide a physical server into multiple VMs.

In real-world scenarios, this virtualization often happens in data centers. Data centers manage physical servers and logically divide them using virtualization software, creating what are called virtual servers, which serve various services.

Before cloud computing became mainstream, companies would use on-premises services—meaning they maintained their own servers. This approach put the burden on the company to manage everything, from power to cooling systems, which was often a struggle.

Seeing the challenges companies faced with on-premises solutions, large companies like AWS, Google, IBM, and Microsoft introduced cloud services. These companies manage everything, and users only pay for what they use. Cloud services have become increasingly popular as a result.

To ensure data is isolated and secure, cloud providers offer services like Virtual Private Cloud (VPC), which allow companies to control access to their resources.

Here's how a typical network flow works in the cloud:

1. When a user searches for a website (e.g., www.example.com), traffic reaches the Internet Gateway.
2. The traffic then goes to the Route Table, which routes it based on request and access rules.
3. Finally, the traffic is sent to the appropriate subnets—public or private—depending on the network configuration.

In networking, servers are identified by IP addresses, such as IPv4 or IPv6. IPv4 addresses consist of four decimal numbers. For example, in the IP address 10.81.0.0/16, the "/16" indicates that 16 bits are used for the network portion, and the remaining 16 bits are for host addresses, giving 65,536 possible IP addresses. This is known as a CIDR block.

To properly configure a VPC, you need to:

1. **Create a VPC** - VPCs are region-specific. AWS provides a default VPC, but you can also create your own through the console by providing a name, IP version, and CIDR block.
2. **Set Up an Internet Gateway** - After creating the VPC, create an Internet Gateway and attach it to your VPC. This can be done via the AWS CLI as well.
3. **Create Subnets** - Create subnets within your VPC, specifying the availability zones and CIDR blocks. Each subnet typically has 256 IP addresses.
4. **Create a Route Table** - Route Tables manage the routing of traffic within your VPC. By default, all subnets are private. Linking a subnet with an Internet Gateway can make it public.
5. **Associate Subnets with the Route Table** - Edit the routes in your Route Table and associate the appropriate subnets to allow traffic as needed.

After setting up your VPC, you can launch applications within the subnets. Data can be exposed based on subnet configurations—public subnets allow traffic, while private subnets are isolated.

When launching instances, make sure to select the appropriate subnet within your VPC. Public instances can be accessed externally, while private instances remain isolated. To connect to private instances, you usually need to use a public instance as a bastion host, connecting via a PEM key.

If you need to install software on a private instance that requires internet access, you will need to set up a NAT Gateway.

This setup ensures your VPC is properly configured for your needs, balancing security and accessibility.